Memorial Sloan Kettering Cancer Center

Using metafeature clustering to mine tissue-specific signals from rare variants in the cancer genome

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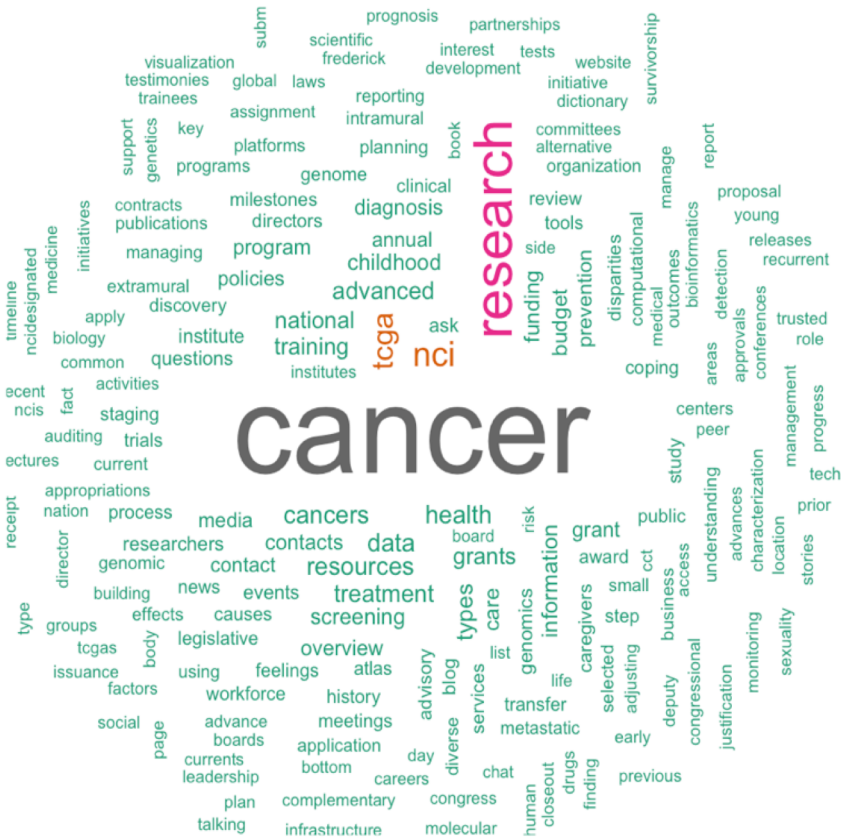
Ethan Ashby, Ronglai Shen, PhD, Saptarshi Chakraborty, PhD

Department of Epidemiology & Biostatistics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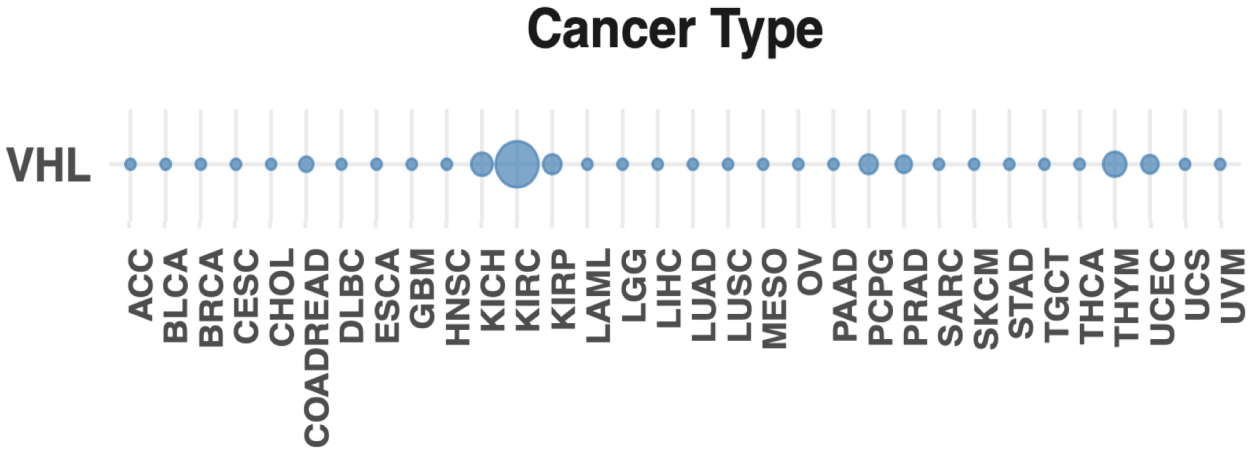
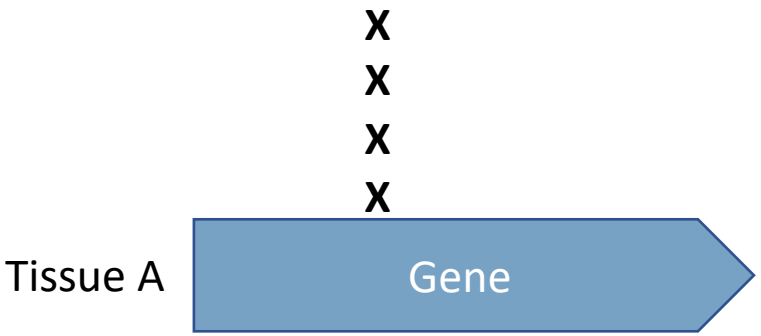
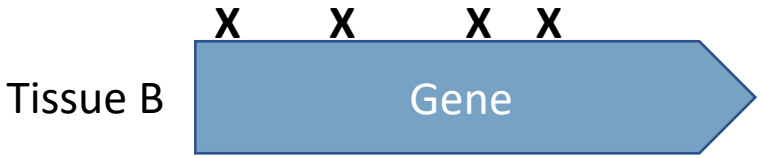
Unseen somatic variant probabilities encode clinically-relevant signals

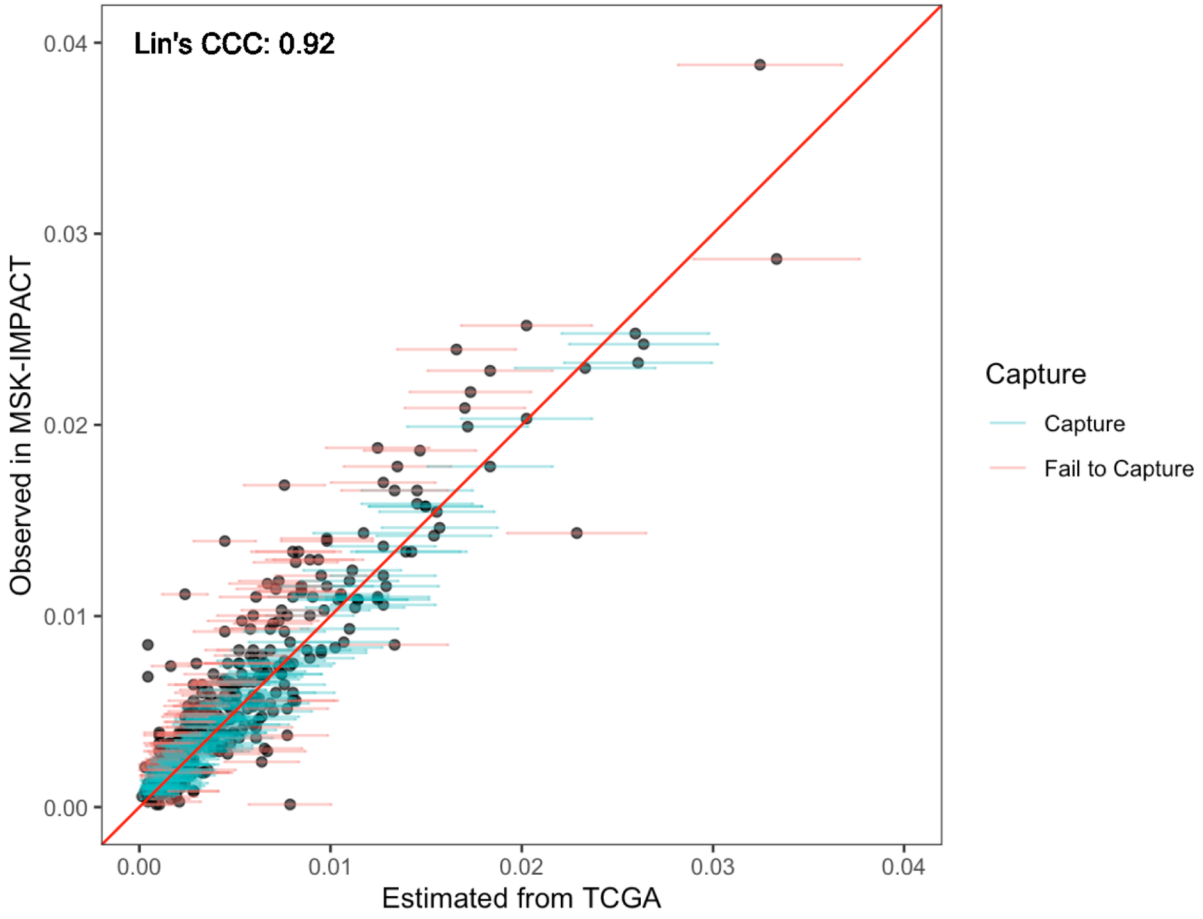
Smoothed Good-Turing Probabilities:
P(X |Tiss. A)< P(X |Tiss. B)







Unseen variant probabilities show strong concordance with validation set



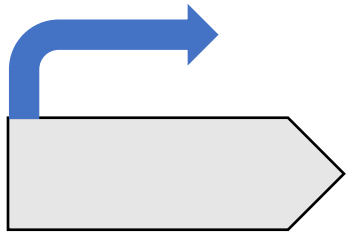
Challenge: sparse mutation data

Strategy: zoom out on the genomic landscape; consider **gene groups or metagenes.**

Goal: identify gene groups with high tissue-type specificity in unseen variant probabilities

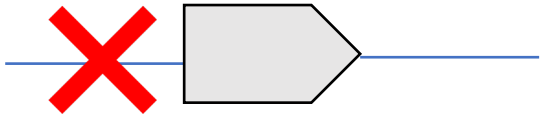
Grouping genes into metagenes could resolve data sparsity challenges and yield tissue-specific signals

Assign ~400 known cancer-related genes to metagenes using 5 correlates of somatic mutation rate



Expression

Replication time



Non-coding mutation rate

GC

GC content

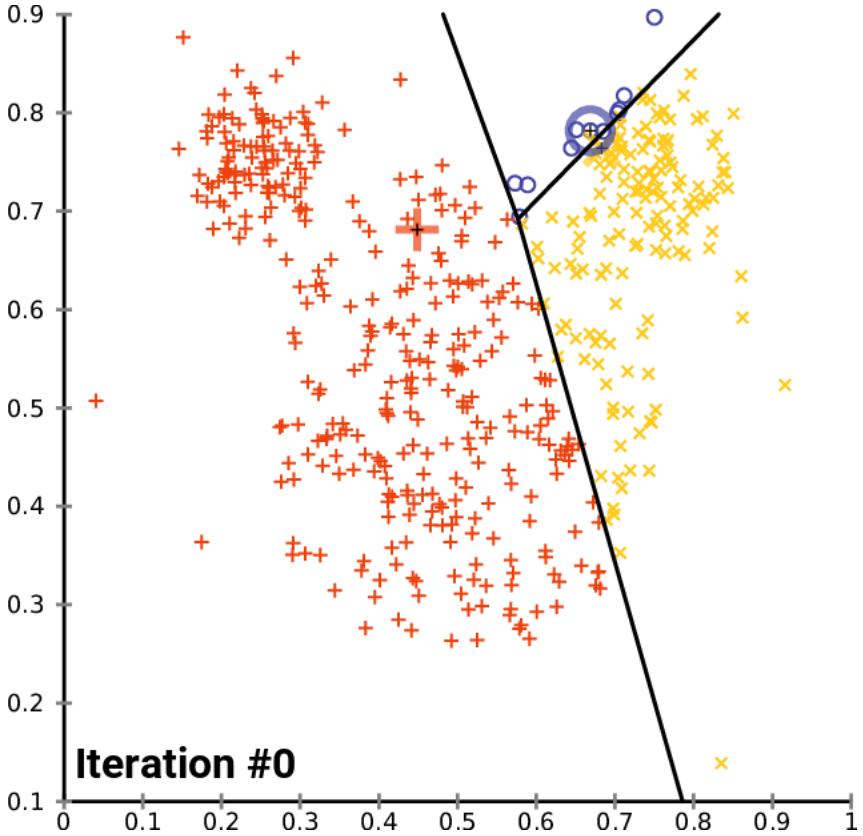
Vs

Open/closed compartment

Idea: gene groups will share similar variable values, similar mutational profiles, and similar unseen variant probabilities

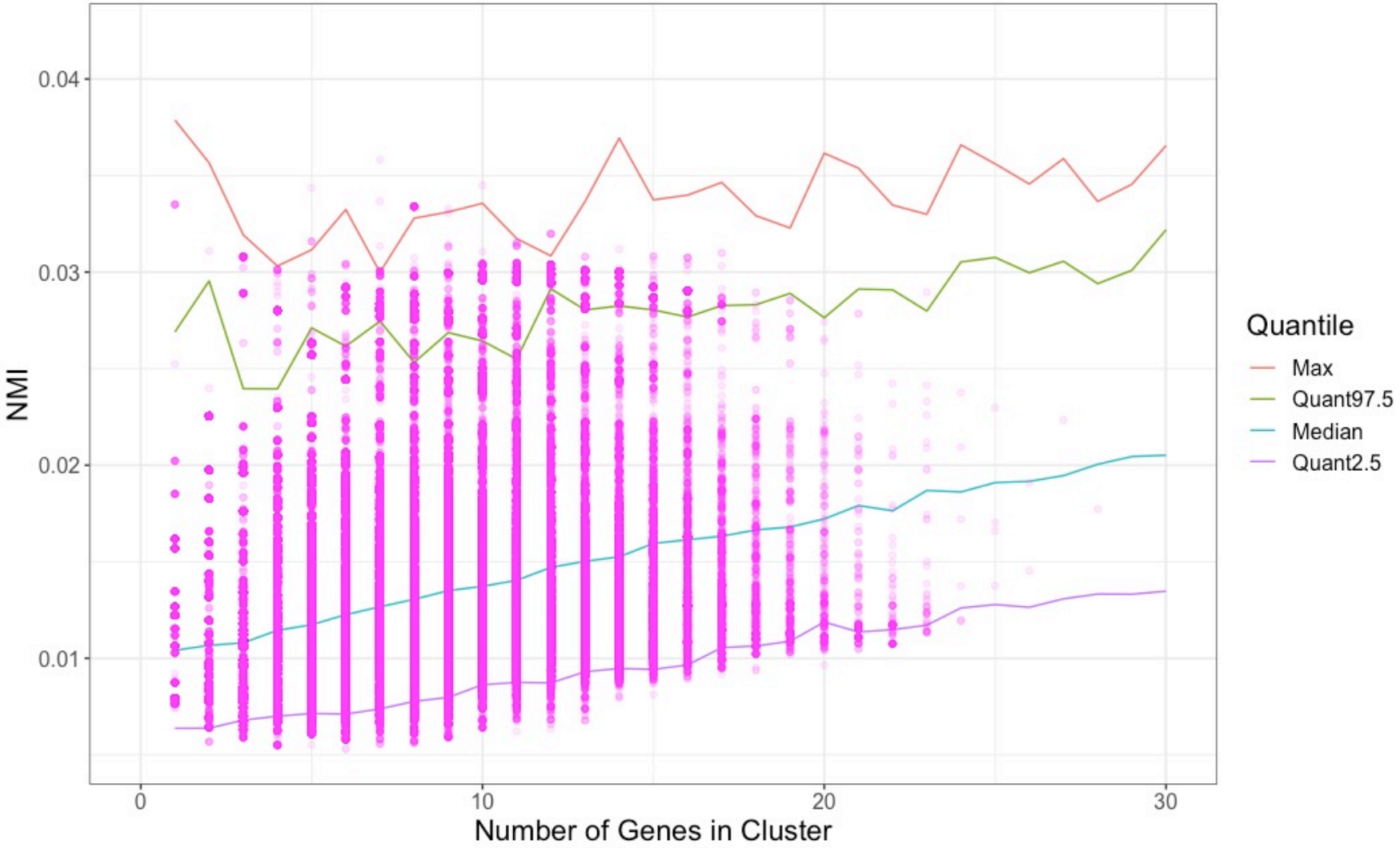
Repeated runs of k-means with random initializations used to generate gene groups

X 10,000



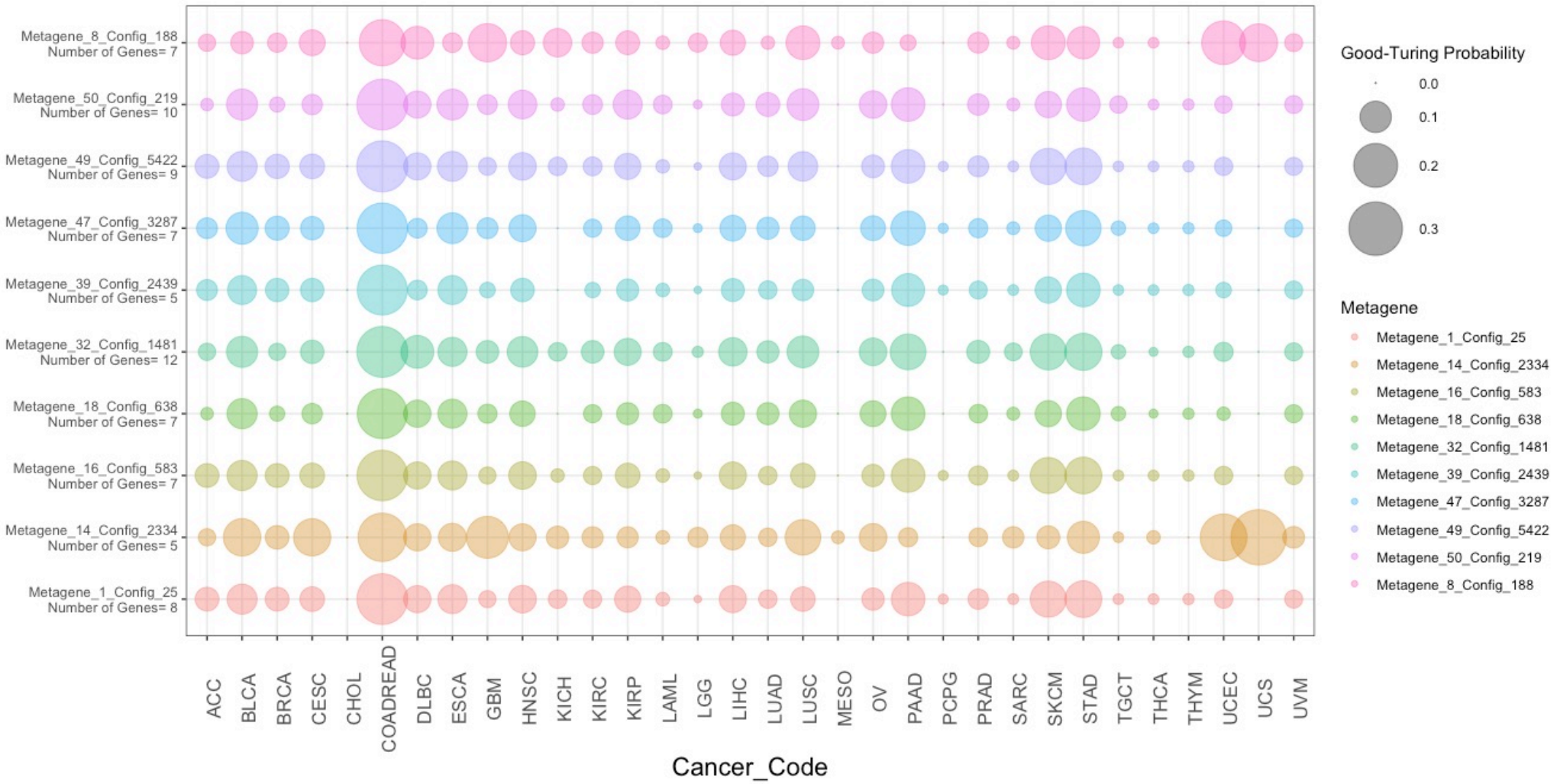
Clustering approach doesn't appreciably improve tissue-specificity compared to random groupings

Number of clusters (k)=50
10,000 k-means clusters

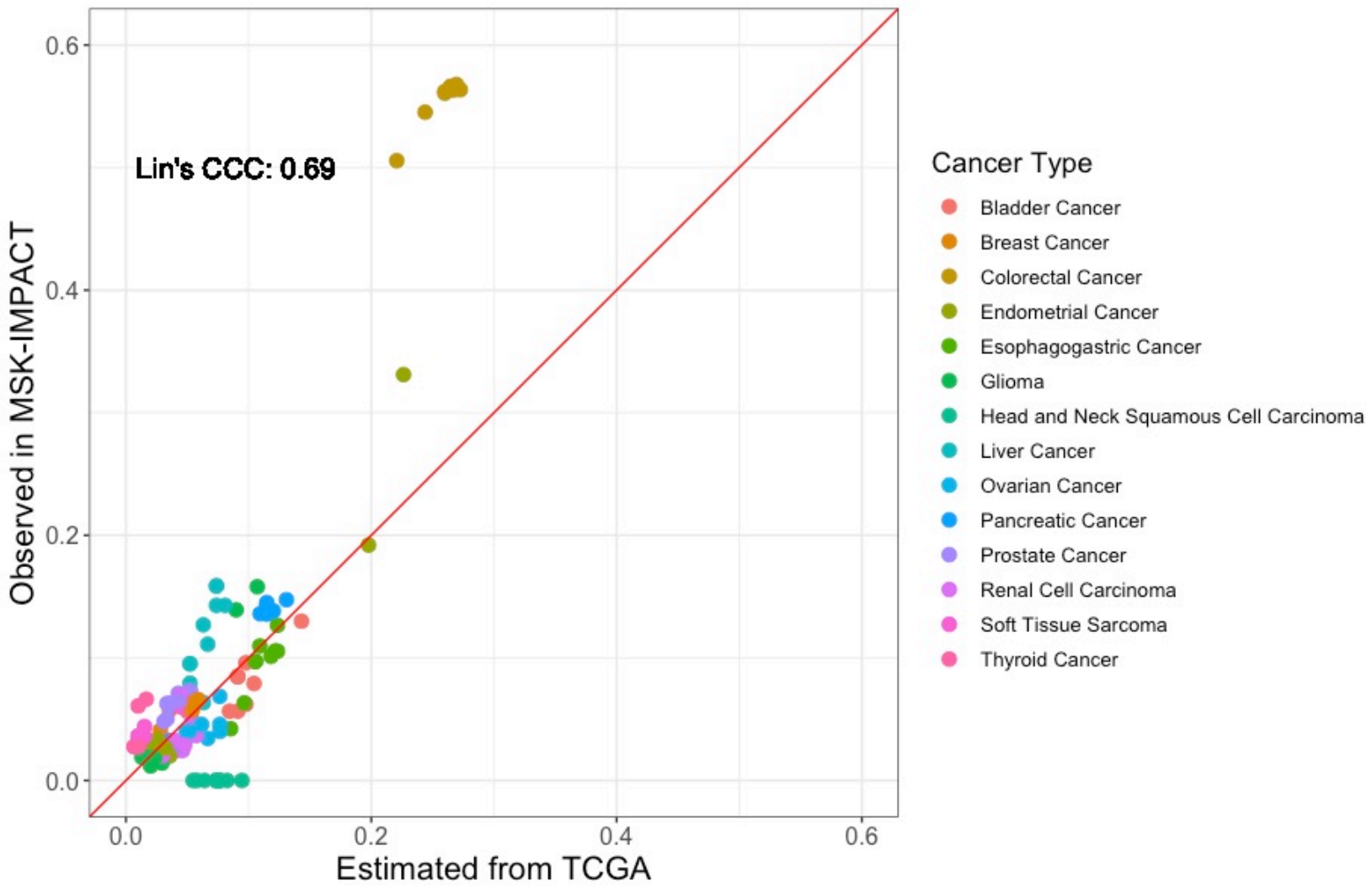


Clustered metagenes show tissue-specific signal

k=50, 10 unique configs



Validating tissue-specific signals in the MSK-IMPACT cohort show agreement in some tissue types, bias in others



Metafeature clustering is a promising approach that requires more work

- **Punchline:** we've developed a framework to resolve mutation data sparseness and yield tissue-specific signals by grouping genes using metafeatures.
- **Next up:** our metafeatures are too crude to significantly improve upon a random, grid-search approach. We need to sophisticate our metafeature space.
- **Down the road:** ID highly tissue specific gene groups and use them to build classifier of primary site

Acknowledgements

- Dr. Ronglai Shen and Dr. Saptarshi Chakraborty
- Dr. Kay-See Tan and Dr. Margaret Du
- My fellow QSURE scholars

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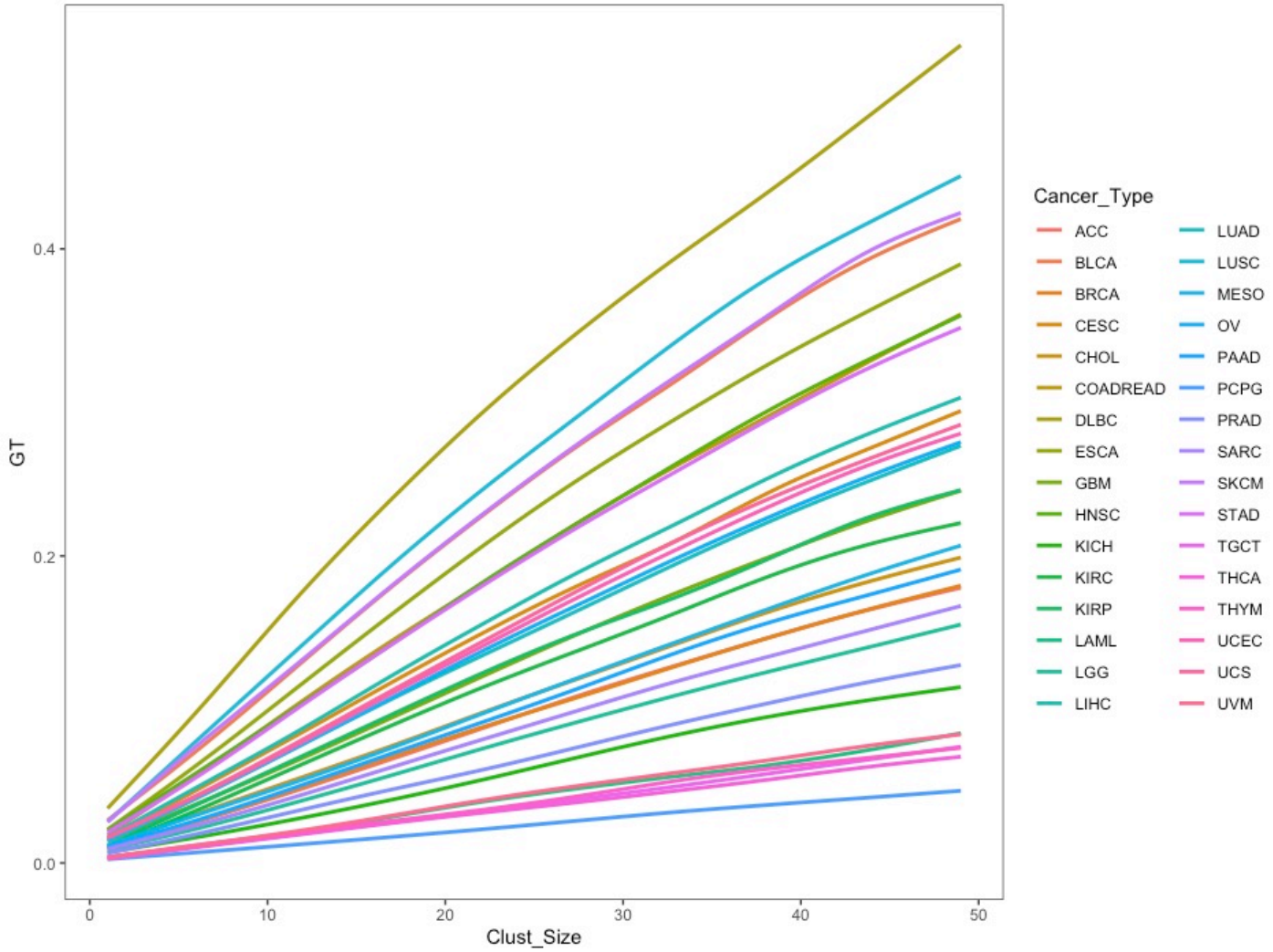
Any questions?

Work Cited

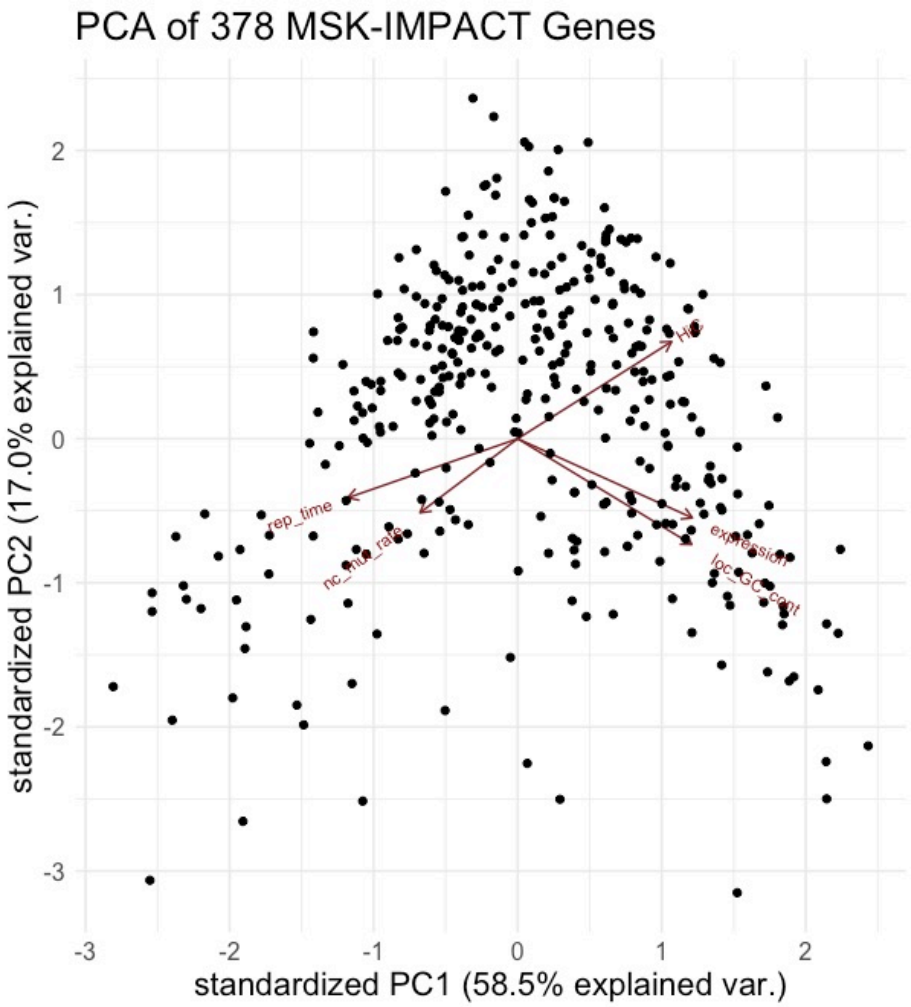
1. Chakraborty, S., Arora, A., Begg, C. B. & Shen, R. Using somatic variant richness to mine signals from rare variants in the cancer genome. *Nat. Commun.* **10**, 1–9 (2019).
2. Gale, W. A. & Sampson, G. Good-turing frequency estimation without tears. *J. Quant. Linguist*. **2**, 217–237 (1995).
3. Ellrott, K. *et al.* Scalable Open Science Approach for Mutation Calling of Tumor Exomes Using Multiple Genomic Article Scalable Open Science Approach for Mutation Calling of Tumor Exomes Using Multiple Genomic Pipelines. *Cell Syst.* **6**, 271–281 (2018).
4. Lawrence, M. S. *et al.* Mutational heterogeneity in cancer and the search for new cancer-associated genes. *Nature* **499**, 214–218 (2013).
5. Zehir, A. *et al.* Mutational landscape of metastatic cancer revealed from prospective clinical sequencing of 10,000 patients. *Nat. Med.* **23**, (2017).

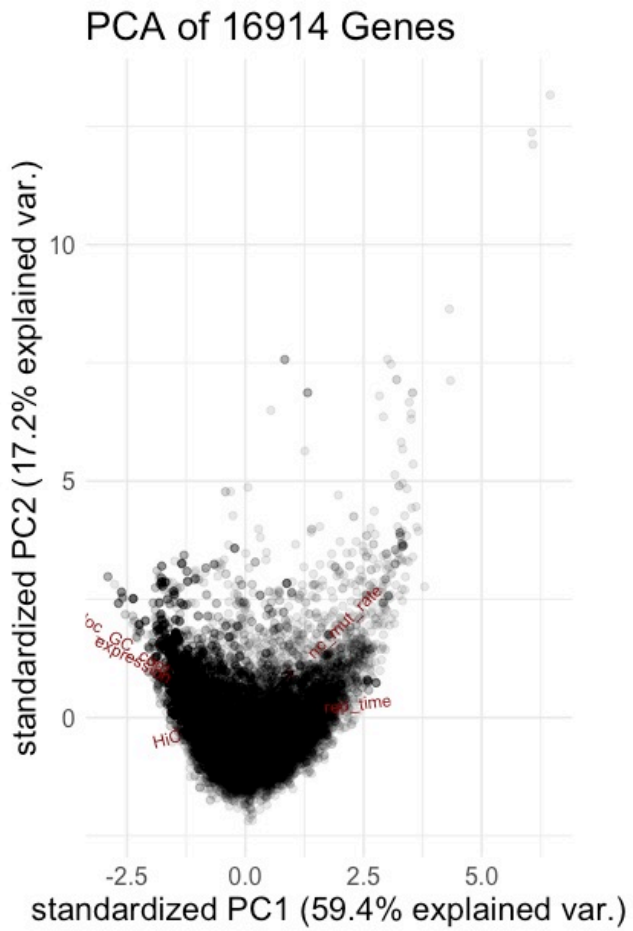
Extra slides

As clusters increase in size, tissue specific probabilities track total mutational burden



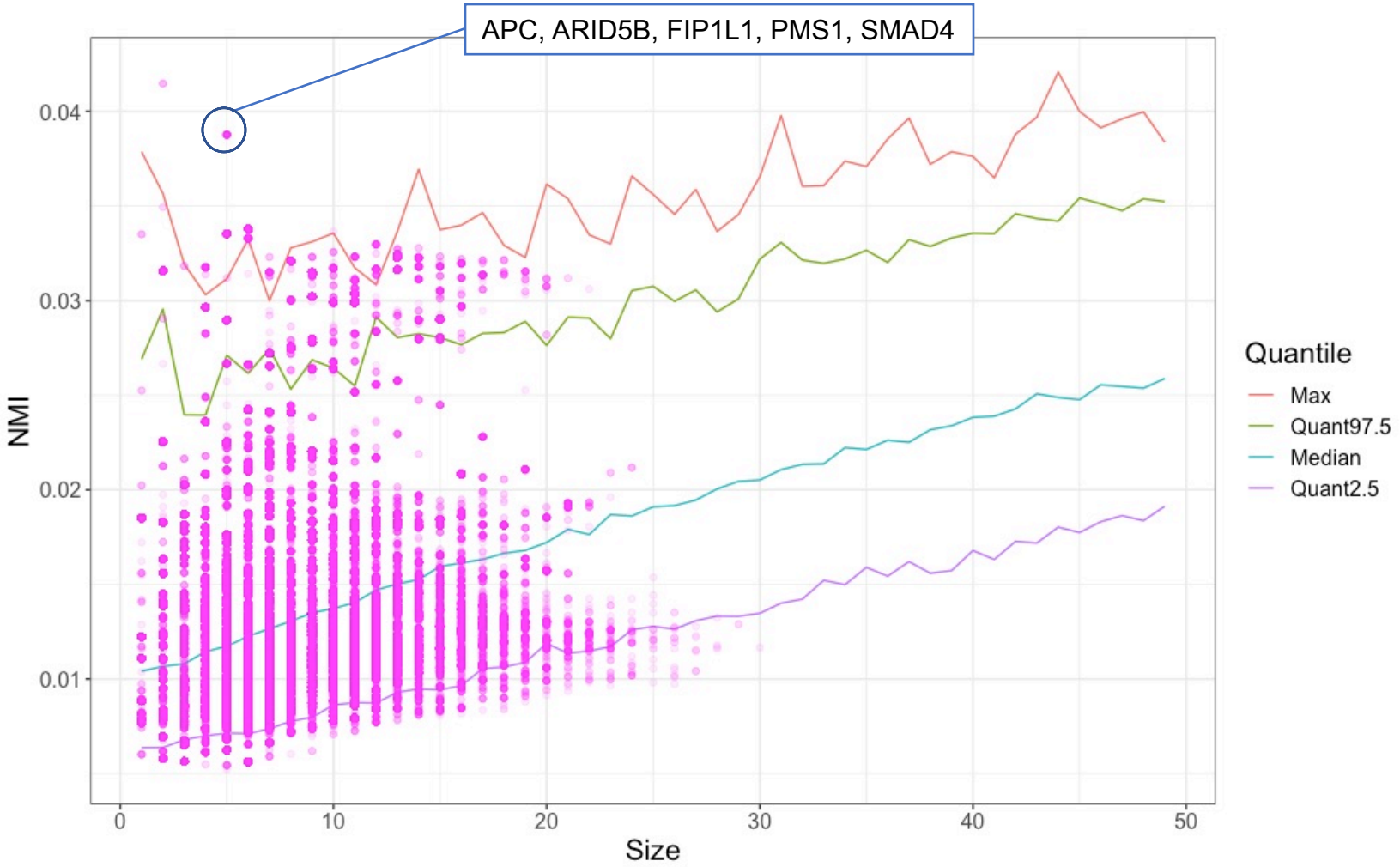
PCA shows weak clustering structure and need for multiple k-means configs





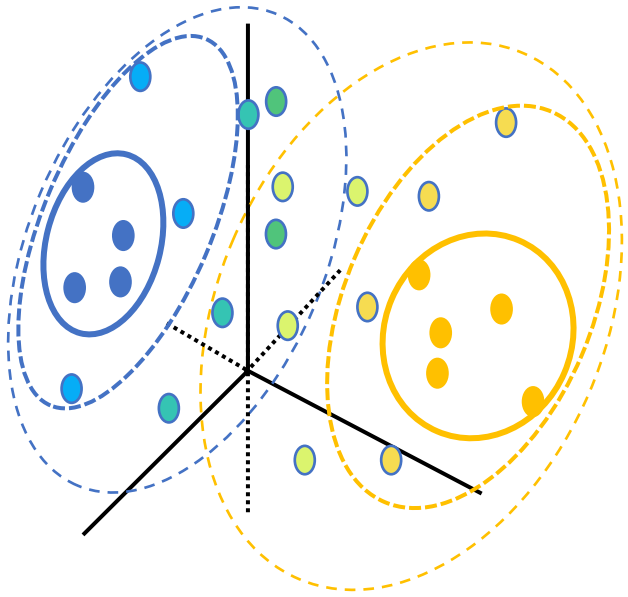
Using principal components to group metagenes may afford signal boost

5000 configurations
k=50
# PC=2



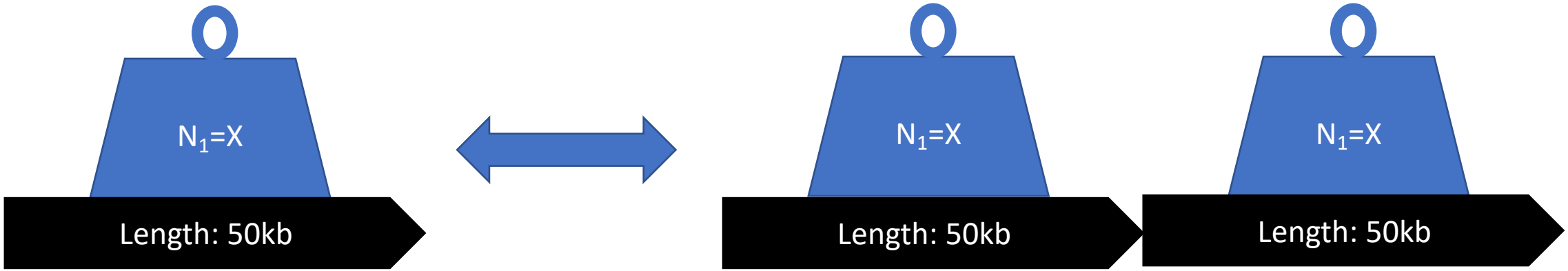
Next steps involve sophisticating metafeatures and loss function

- Improvements to clustering approach
  - Weighting of clustering variables
  - Feature selection
  - Learn optimal cluster number and soft clustering using GMM
  - Expand clustering variables
    - Incorporation of more transcriptomic metafeatures: average gene expression for 32 tissue types
- Incorporate NMI into the clustering loss function

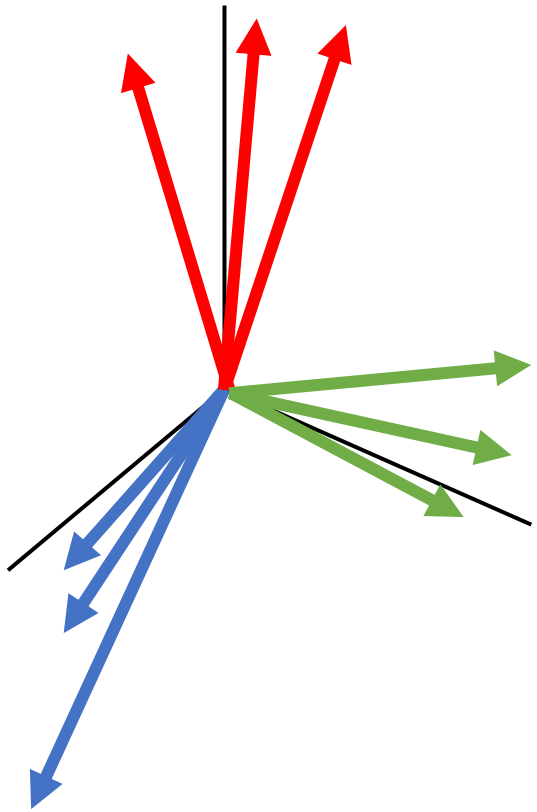


We standardized the variant frequencies per 50kb exonic length

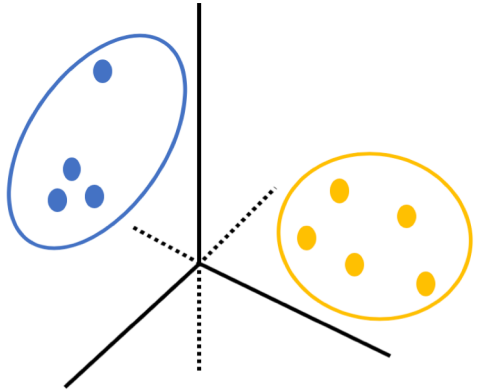
- Good-Turing formula: $\mathrm{P}(\geq 1 \text{ unseen variant}) = 1 - e^{\frac{-N_1}{m+1}}$
- Good-Turing formula Scaled: $\mathrm{P}(\geq 1 \text{ unseen variant}) = 1 - e^{\frac{-N_1}{m+1} * S}$ where $S = \frac{scaling\ factor}{feature\ length}$



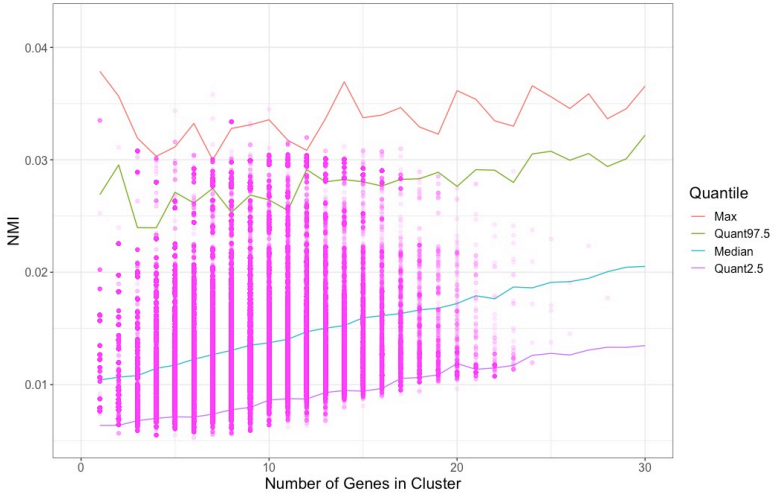
Cluster genes based on cosine distance



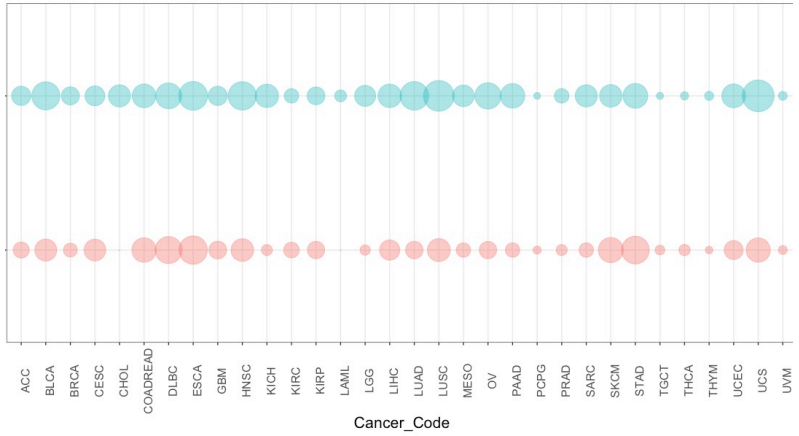
Grouping genes into metagene clusters could resolve data sparsity challenges and produce tissue-specific signals



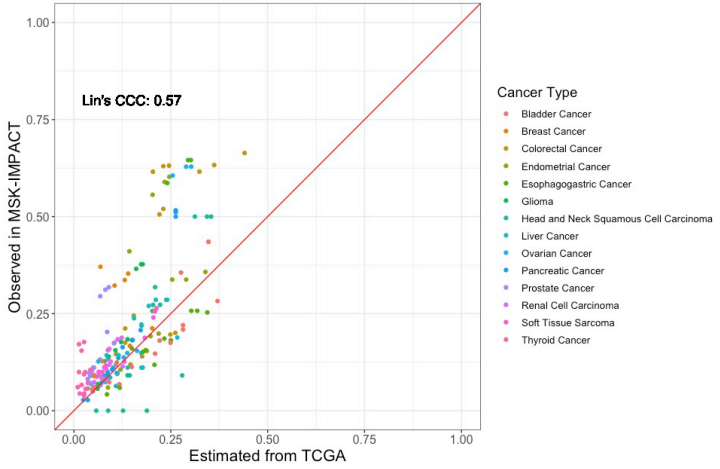
Learn metagenes from group of ~400 genes using metafeatures x 10,000



Create null (random) gene groups and select cluster-based metagenes with significant NMI relative to null

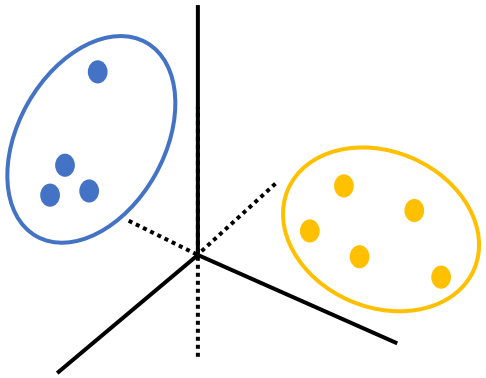


Create probability "fingerprint" for each metagene with TCGA data
Assess specificity of the fingerprint using NMI



Validate fingerprints on MSK-IMPACT data

Repeated runs of k-means with random initializations used to generate gene groups

X 10,000

1. Choose random cluster centers
2. Assign each data point to its closest center
3. Update the centers by taking mean coordinates of all cluster members